and that the proposal would unfairly discriminate against retail investors.[357] One commenter suggested that the order protection rule fosters retail investor confidence and that not protecting the new round lots could erode trust in the markets.[358] Another commenter suggested that market makers may not protect round lot orders that are not within the scope of the order protection rule and stated that limiting order protection to transactions of 100 shares or more would "discourage limit orders – a valuable driver of price discovery – as investors would be less likely to receive execution without price protection."[359]

Separate Rulemaking: Some commenters stated that they disagreed with the proposal to separate order protection from the NBBO, and if the Commission were to make such a change, it should do so in a separate rulemaking.[360]

The Commission is not adopting the proposed amendment to the definition of protected bid or protected offer.[361] A wide range of commenters expressed significant concerns regarding this aspect of the proposal, including concerns about complexity and ambiguity that could stem from the best execution, routing, and order handling ramifications of introducing round lot quotes that are unprotected, a potential increase in trade-throughs and locked and crossed markets, and possible erosion of confidence among retail investors that their orders are being treated fairly. The Commission recognizes these concerns and does not believe that adopting the

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357 See Nasdaq Letter IV at 16.

358 See TD Ameritrade Letter at 10.

359 Schwab Letter II at 2.

360 See SIFMA Letter at 14; Nasdaq Letter IV at 14–15.

361 However, the Commission is still making the technical change to delete the references to "The Nasdaq Stock Market, Inc." in the definition. The Commission did not receive comments on this piece of the proposal and continues to believe the language is redundant because the Nasdaq Stock Market is now a national securities exchange. See Proposing Release, 85 FR at 16749.

proposed amendments to the definition, at this time, would appropriately advance the broader objectives of the proposal, particularly enhancing the utility and availability of consolidated market data.

In support of its preliminary belief that Rules 611 and 610(d) should not be extended to the smaller-sized quotations reflected in the proposed definition of round lot, the Commission cited concerns expressed by various market participants about the existing scope of these rules.[362] However, given the concerns expressed in comments on the proposal, the Commission believes that not extending Rules 611 and 610(d) to the new round lot sizes could create unnecessary complexity and confusion. For example, the Commission continues to believe that improvements in trading and order routing technology since 2005 and the applicability of best execution requirements to orders of all sizes would incentivize market participants to engage with orders in the new round lot sizes even if they were not protected, and that these technological improvements and market forces would also help mitigate excessive locking or crossing of quotations in the new round lot sizes. However, the Commission also agrees with commenters that consistently applying Rules 611 and 610(d) to all round lot sizes would promote confidence among investors and other market participants that market participants will engage with orders in the new round lot sizes and that orders in the new round lot sizes will not be excessively locked or crossed.

Furthermore, other modifications to the proposal that the Commission is adopting herein mitigate the Commission's concerns, as expressed in the Proposing Release, about the expansion of the order protection requirements in Rule 611 and the prohibitions on locked and crossed markets in Rule 610(d). Specifically, as discussed above, the Commission is: (a) modifying the

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[362] See Proposing Release, 85 FR at 16747–49.

proposed definition of round lot so that only stocks priced over $250.00 will be assigned to a round lot size less than 100; and (b) increasing the notional size thresholds of the new round lot sizes so that protected order interest at the new round lot sizes is more meaningful.[363] Further, since currently only 134 stocks are priced over $250.00,[364] the scope of the extension of Rules 611 and 610(d) would be fairly limited, extending to a much smaller number of stocks than under the proposed amendments.[365] Furthermore, of this number, six NMS stocks already have a round lot size of less than 100 today.[366] Therefore, only 128 NMS stocks would receive trade-through protection for smaller-sized orders. Additionally, with the larger notional size of $10,000 adopted for the definition of round lot,[367] the extension of order protection to round lots would apply only to orders of a more substantial size.

(b) Odd-Lot Aggregation for Protected Quotations

A number of commenters discussed odd-lot aggregation with respect to protected quotes, and most stated that the Commission should allow aggregation across multiple price levels, instead of at just one level.[368] Some commenters stated that preventing odd-lot orders from being aggregated across different price levels to create protected quotes, as proposed, would cause spreads to widen, with one commenter providing empirical data, based on quoting activity

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363 See supra Section II.D.

364 This information is based on data analyzed by staff for September 2020.

365 By comparison, under the proposed definition of round lot, Rules 611 and 610(d) of Regulation NMS would have extended to over 1,600 stocks in the absence of the proposed amendment to the definition of protected bid or offer.

366 See infra Section II.E.2(c).

367 See supra Section II.D.

368 See, e.g., AHSAT Letter at 4.

during the month of April 2020, showing that average spreads would widen by over $0.50 for shares with prices of $500.01 to $1,000.00 and nearly $1.50 for shares with prices of $1,000.01 or more.[369] One commenter suggested this restriction would hurt retail investors because it would result in their orders being executed at inferior prices.[370]

Another commenter suggested that only allowing odd-lot aggregation at one price level, instead of across price levels, would lead to fewer protected quotes.[371]

One commenter suggested that different odd-lot aggregation methodologies between the NBBO and the PBBO would create confusion among market participants.[372] That commenter also suggested that different methodologies could raise confusion from a best execution perspective.[373] Similarly, one commenter suggested only allowing odd-lot aggregation at one price level, combined with the proposal to provide all levels of depth up to the PBBO, would "create[] implementation difficulties."[374]

Another commenter suggested that, given the proposal's intent to maintain the status quo with respect to the scope of orders that are subject to Rule 611, the Commission should consider

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369 See Cboe Letter at 9–12; IEX Letter at 7; Nasdaq Letter IV at 16–17; Virtu Letter at 5.

370 See Cboe Letter at 10–11.

371 See TD Ameritrade Letter at 11.

372 See BlackRock Letter at 4; SIFMA Letter at 7–8 (suggesting different odd-lot aggregation methodologies would "rais[e] additional technical issues" and recommending that the Commission should provide clarification as to how depth-of-book data is determined, "particularly because the aggregation process appears to work differently for the PBBO versus the NBBO, BBO and depth-of-book determinations").

373 See BlackRock Letter at 4.

374 MEMX Letter at 4.

continuing the existing market practice, as codified in exchange rules, to aggregate quotes at multiple price levels.[375]

One commenter stated that if order protection were extended to all round lots, the commenter would "support aggregating odd lots in the manner currently described for the PBBO."[376]

In light of these comments, the Commission is modifying the proposed definition of core data to require odd-lot aggregation across multiple prices for purposes of the NBBO and protected quotations.[377] The Commission believes that this approach will: (a) provide a consistent odd-lot aggregation methodology for all elements of core data; (b) be consistent with existing practices, as some commenters pointed out,[378] and would avoid potential costs associated with changing these practices; and (c) avoid unintended consequences that could adversely affect investors, such as widening spreads or reducing the number of protected quotes. As stated in the Proposing Release,[379] this methodology effectively extends order protection to the aggregated odd-lot orders. However, as discussed above,[380] a majority of commenters who opined on odd-lot aggregation for protected quotations preferred a cross-price methodology. Lastly, the comment raised regarding the need for a consistent odd-lot aggregation methodology

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375 IEX Letter at 7.

376 Capital Group Letter at 4.

377 For example, if there is one 50-share bid at $25.10, one 50-share bid at $25.09, and two 50-share bids at $25.08, the adopted cross-price odd-lot aggregation method would show a protected 100-share bid at $25.09, while the proposed single-price odd-lot aggregation method would show a protected 100-share bid at $25.08.

378 See, e.g., BlackRock Letter at 4.

379 See Proposing Release, 85 FR at 16747–50.

380 See supra notes 368 through 376.

for the NBBO and protected quotations is less relevant in light of changes the Commission is adopting herein—namely, that the new round lot NBBO will be protected.

(c) Removing Protected Status from Certain NMS Stocks

Some commenters stated that removing protected status for those NMS stocks that currently have a protected quotation size of under 100 shares would potentially increase the number of locked and crossed markets for the twelve current NMS stocks that have protected quotations of under 100 shares.[381] One commenter stated that the Commission did not analyze the effects of removing order protection from those twelve stocks.[382]

The Commission understands these concerns regarding the possible reduction of order protection and locked/crossed markets restrictions for these 12 stocks in their current round lot sizes. However, as discussed above,[383] the Commission is not adopting the proposed amendments to the definition of protected bid or offer, so all NMS stocks—including the 12 that currently have non-100 round lot sizes—would be assigned to round lot sizes based on their per share price, and all round lot orders in these stocks would be protected quotations subject to the trade-through prevention requirements of Rule 611 and the locked and crossed markets restrictions of Rule 610(d).

In addition, the Commission acknowledges that, based on staff analysis of stock prices as of September 2020, one stock will have its round lot size increased from 10 to 40, one will have its round lot size increased from 1 to 10, and six will have their round lot size increased from 10

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381 See SIFMA Letter at 13–14.

382 See NYSE Letter II at 6.

383 See supra Section II.E.2(a).

to 100. As a result, order protection pursuant to Rule 611 and the locked/crossed markets prohibitions of Rule 610(d) will no longer apply to some smaller orders in these stocks.[384] However, the Commission continues to believe that determining the round lot sizes of all stocks based upon price, without special exceptions for certain stocks that currently have non-standard round lot sizes, will reduce complexity and implementation costs, set consistent expectations regarding round lot sizes among market participants, facilitate the transition to price-based round lots, and justify any potential costs of increasing the round lot sizes of a limited number of stocks.[385]

F. Definition of "Depth of Book Data" under Rule 600(b)(26)

1. Proposal

The Commission proposed to include "depth of book data," defined as follows, as an element of core data: all quotation sizes at each national securities exchange, aggregated at each price at which there is a bid or offer that is lower than the best bid down to the protected bid and higher than the best offer up to the protected offer, and all quotation sizes at each national securities exchange, aggregated at each of the next five prices at which there is a bid that is lower than the protected bid and offer that is higher than the protected offer.

The Commission solicited comment on various aspects of the proposed definition of depth of book data, including whether the definition captures the appropriate level of depth data and whether the Commission should include more or fewer levels of depth or otherwise revise the definition to capture the key depth information that would be useful to market participants.

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384 In addition, for four of the twelve stocks that currently have non-100 share round lot sizes, the round lot size would not change, and these stocks would not experience a change in terms of the applicability of Rules 611 or 610(d) of Regulation NMS.

385 See Proposing Release, 85 FR at 16749.

2. Final Rule and Response to Comments

The Commission received comments expressing support for the proposed definition of depth of book data and comments raising various concerns, recommendations, and requests for clarification. As discussed in more detail below, the Commission is adopting a modified definition of depth of book data in response to comments.

(a) Support for Five Levels of Depth

Several commenters expressed general support for including depth of book data in core data[386] or specifically agreed with the Commission's proposed five levels of depth,[387] with some explaining that five levels of depth is suitable for certain market participants or trading

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386 See Angel Letter at 1–9 (stating that including depth of book in core data is a "great idea" and emphasizing the importance of depth of book data, particularly to retail investors trading in illiquid stocks of smaller companies and using liquidity-providing limit orders); letter from Allison Bishop, President, Proof Trading, to Vanessa Countryman, Secretary, Commission, dated May 1, 2020, ("Proof Trading Letter") at 1; Healthy Markets Letter at 3; DOJ Letter at 2–4; NovaSparks Letter at 1.

387 See IEX Letter at 5; Capital Group Letter at 2–3; Fidelity Letter at 4–5; Schwab Letter at 4; T. Rowe Price Letter at 2; Wellington Letter at 1; Virtu Letter at 5; SIFMA Letter at 7; STANY Letter II at 4; IntelligentCross Letter at 4. In addition, staff conducted the same analysis of book depth that was included in the Proposing Release but evaluated proprietary market data from a more recent time period (the week of May 4, 2020) than was included in the Proposing Release (July 19, 2019). The results were very similar and confirmed the Commission's view that there is a substantial amount of quotation volume several levels below the best bid. Staff observed substantial quotation volume several levels below the best bid (the offer side was not examined). On average, there is quotation interest at every $0.01 increment at least ten levels out for the most liquid stocks; for the least liquid stocks, there is a large gap between the best bid and the next highest bid, and large gaps are generally also present between the next several bid levels. In addition, the staff review found a significant percentage of the total notional value of all depth of book quotations for both liquid and illiquid stocks falls within the first five price levels.

practices.[388] One commenter expressed general support for the expansion of core data, including depth of book data, provided that it does not materially increase overall latency.[389]

The Commission agrees that including depth of book data in core data will be useful for and beneficial to a variety of market participants and is adopting the definition of depth of book data largely as proposed, with certain modifications.[390] The Commission does not believe that including depth of book data in core data will materially increase the overall latency of core data for several reasons. First, only five levels of depth are included, limiting the number of quotes necessary to be processed.[391] Moreover, the decentralized consolidation model will result in consolidated market data, including depth of book data, being delivered to market participants with lower latencies.[392] In addition, the Commission believes that market participants that choose to receive the full set of consolidated market data, including depth of book data, will either leverage the existing technology that they currently use to receive depth of book data on a

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388 See State Street Letter at 2–3 (stating that institutional firms generally use up to five levels of depth for order routing); ICI Letter at 8–9 (stating that depth of book data "should consist of at least five price levels, based on the typical trading needs of funds"); Clearpool Letter at 14 (stating that five levels of depth of book data is sufficient to improve the usefulness of core data for most market participants and that "five price levels typically tend to be a sufficient level of depth for Clearpool for sweeping multiple levels of the book in executing an order"); SIFMA Letter at 7 ("[A] review of our institutional member firms found that while some used less than five levels and others used more, five levels of depth strikes an appropriate balance for the order routing purposes of most.").

389 See Citadel Letter at 1.

390 See infra Sections II.F.2(b) through II.F.2(c).

391 See infra Section II.F.2(d) (discussing comments suggesting that full, order by order depth of book should be included in core data and the potential latency ramifications thereof).

392 See infra Section III.B.5.

proprietary basis or make the investments in technology to receive the data so that the additional content will not add significant latency.[393]

(b) Proposed Definition Would Include More than Five Levels of Depth

Some commenters stated that the proposed definition would include five levels of depth on each exchange, which could be confusing and costly and could add latency by increasing actual quote traffic and information to be processed significantly beyond five levels.[394] Some commenters also stated that the proposal to include price levels between the best and protected quotes could similarly include a large number of additional data points.[395] Some commenters recommended that the Commission clarify how depth of book data will be determined and made available on an individual exchange basis, particularly with respect to how odd-lots would be aggregated at depth of book price levels.[396]

The Commission believes that these comments highlight the need to more clearly articulate the scope of data that the definition of depth of book data includes and how this data

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393 See infra Section V.C.1(c)(iv).

394 See STANY Letter II at 4; TD Ameritrade Letter at 5 ("[T]he proposal to include five levels of depth would add ten quotes per security for every exchange, which today amounts to 130 new data points per security.").

395 See STANY Letter II at 4; TD Ameritrade Letter at 5 ("[T]the Proposal includes all price levels between the BBO and the protected quote, which for higher priced securities could add hundreds more data points."); NYSE Letter II at 5 (stating that the number of price levels between the best and protected quotes, as proposed, could be more than five levels and could fluctuate intra-day as quotes update).

396 See SIFMA Letter at 7.